



AI-Powered Real-Time Bidirectional Translation Between Turkish Sign Language and Turkish



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Signa Turk is an AI-powered, real-time, bidirectional communication system designed to reduce barriers between deaf and hearing individuals. The platform translates Turkish Sign Language (TSL) into spoken Turkish and converts spoken Turkish into sign-based avatar animation. By combining computer vision, speech recognition, language modeling, and real-time animation, Signa Turk provides an accessible and interactive digital translation experience.



- Communication barriers still limit accessibility in daily and educational environments.
- Existing systems are often one-directional or limited to isolated signs.
- A practical bidirectional interface is needed for real-time inclusive communication.



- Translate TSL signs into meaningful Turkish speech.
- Convert spoken Turkish into sign-language animation.
- Provide a real-time, user-friendly, bidirectional interaction interface.
- Build a reusable digital TSL animation library for future development.

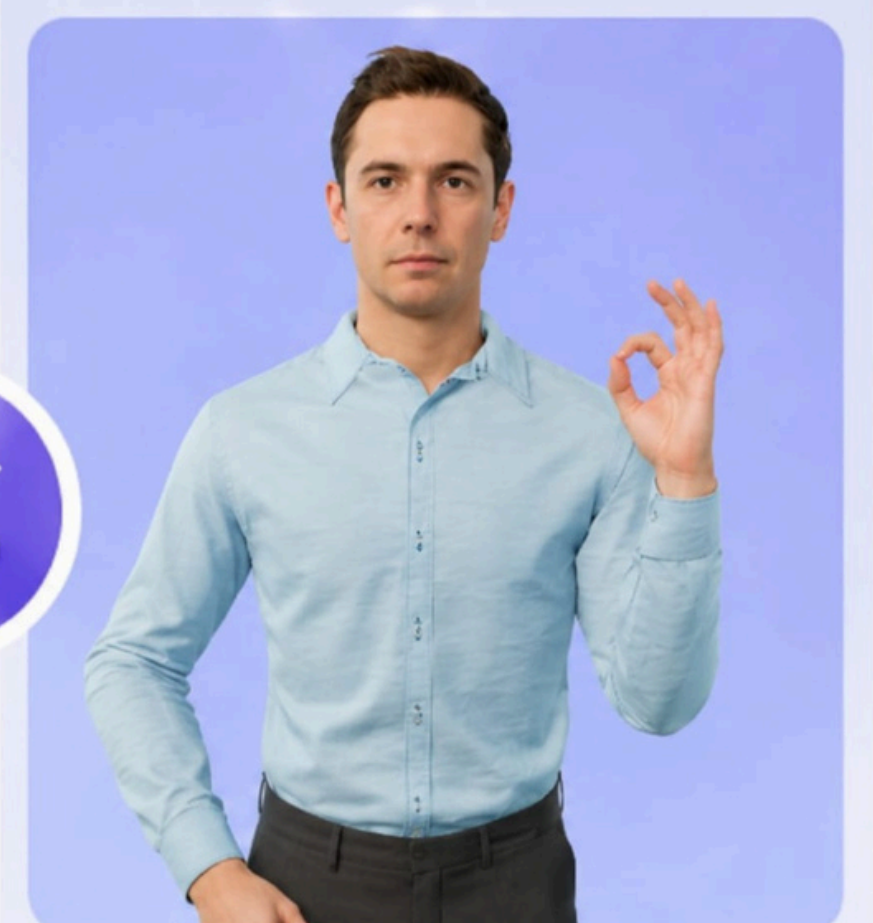
REAL-TIME BIDIRECTIONAL INTERACTION

YOU SIGN



Detected: EV

AVATAR RESPONDS



 Evet!

SYSTEM PIPELINE

SIGN TO SPEECH

- **1. Camera/MediaPipe**
Capture video and detect hands using MediaPipe.
- **2. Hand Detection & Crop**
Detect hands and crop bounding box.
- **3. CNN/BiLSTM Model**
Deep learning model for sign (letter/word) recognition.
- **4. Letter or Word Classification**
Recognize the sign and classify as letter or word.
- **5. GPT LLM (Context-Aware)**
Combine recognized tokens into context-aware sentences.
- **6. TTS Output (Turkish)**
Convert sentence to natural Turkish speech.

SPEECH TO SIGN

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- 1. Microphone Input**
Capture user's spoken Turkish.
 - 2. Whisper Speech-to-Text**
Transcribe speech using Whisper (medium, TR).
 - 3. Text Parsing**
Parse text and segment into words/letters.
 - 4. Unreal Engine + MetaHuman**
Render sign animations with realistic MetaHuman avatar.
 - 5. Animation Player**
Play pre-rendered animations sequentially.
 - 6. Word MP4s (Fallback)**
Use letter-level MP4s for unknown words.



MODELS & PERFORMANCE

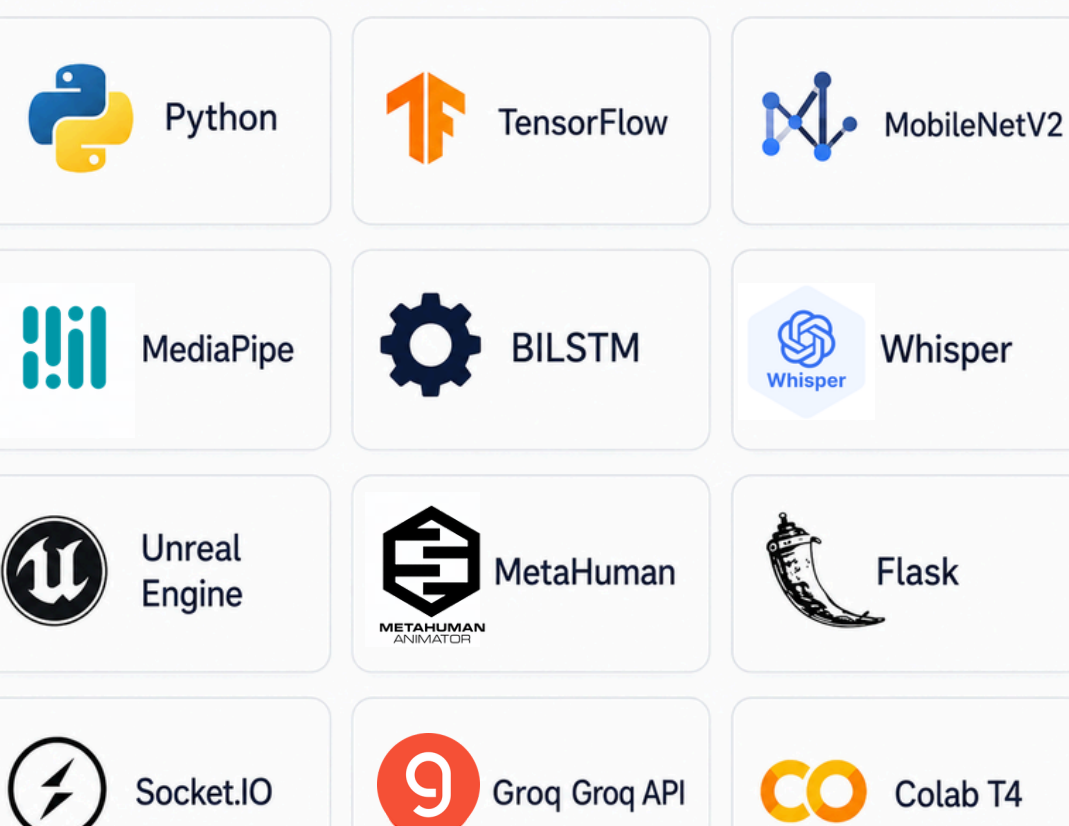
Model/Component		Accuracy
CNN Letter Model (MobileNetV2, fine-tuned)		99.4%
BiLSTM Word Model		92.1%
MediaPipe	Hand landmark detection	
Whisper (medium, TR)	Speech recognition	
Groq LLM	Context-aware sentence assembly	

DATA COLLECTION & CAPTURE REQUIREMENTS

- ✔ **Lighting:** Avoid direct sunlight or harsh highlights; balanced indoor lighting is ideal.
- ✔ **Hand position:** Signs should be performed clearly within the camera frame.
- ✔ **Distance:** Approximately 50-120 cm from the camera.
- ✔ **Background:** Plain, contrasting background improves detection accuracy.
- ✔ **Sign duration:** Each sign should remain visible for at least 8 frames (-0.3 s) for stable confirmation.

ORIGINAL VALUE
/CONTRIBUTIONS

- ✓ First bidirectional Turkish digital interpreter concept combining sign-to-speech and speech-to-sign in one interface.
- ✓ Hybrid recognition approach: CNN for letters + BiLSTM for words.
- ✓ Real-time chat-style interface built with Flask and SocketIO.
- ✓ Creation of a standardized TSL animation library for digital accessibility.
- ✓ Integration of AI, speech technologies, and avatar animation in a unified pipeline.

 TECH STACK

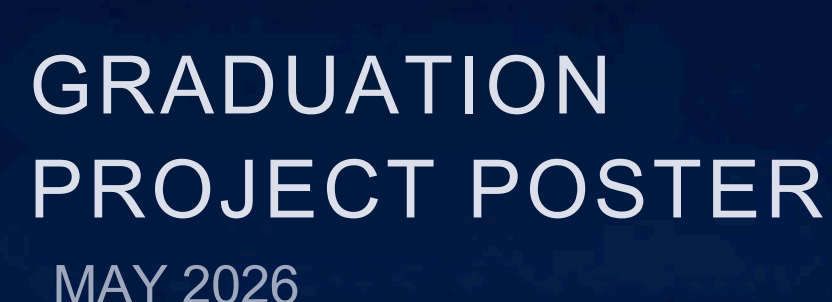
RESULTS & IMPACT

The developed system demonstrates that a real-time bidirectional TSL-Turkish communication platform is feasible and effective. Experimental results show high letter-recognition accuracy and strong word-level performance, while the avatar-based output improves accessibility and user engagement.

- ✓ Real-time multimodal interaction
- ✓ Improved accessibility for deaf users
- ✓ Scalable architecture for future dataset expansion

FUTURE WORK

- ✓ Increase vocabulary size and sentence coverage.
- ✓ Expand the animation library and facial expressions.
- ✓ Optimize mobile and web deployment.
- ✓ Improve continuous sign recognition.
- ✓ Conduct user testing with larger participant groups.



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